

Puzzle sheet

And/or problems

0.28	0.32	0.675	0.225	0.17	0.02	0.56

0.17	12

0.02	0.84	0.675

12	0.3	0.18	0.55

0.3	8

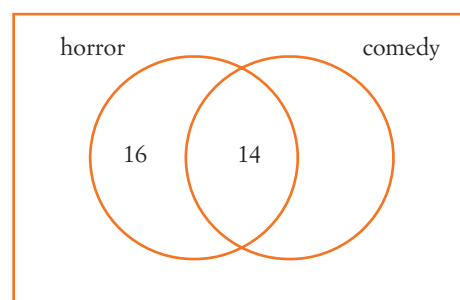
0.025	0.17	0.02

The answer to each question below has a corresponding letter. The numbers in the grid above are the answers to the questions. Calculate the answers to the probability questions. Then fill in the boxes above with the letters that match the answers, to discover a quote from William Shakespeare's *Hamlet*.

Questions

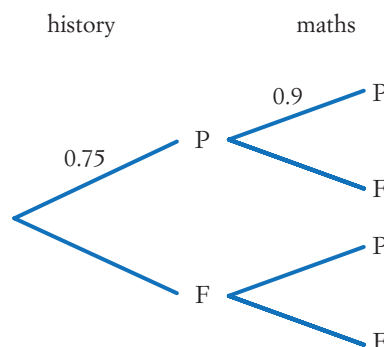
- 1 In a group of 50 teenagers, 30 like horror films, 26 like comedies and 14 like both horror films and comedies. Complete the Venn diagram and then find:

- a the number of teenagers that like comedy but not horror films _____ **S**
- b the number of teenagers that do not like either horror films or comedies _____ **F**
- c the probability a teenager would like horror films or comedies _____ **H**
- d the probability a teenager would like horror films and comedies _____ **B**
- e the probability a teenager likes only horror films. _____ **R**



- 2 Matt has two tests to sit, in History and Mathematics. He estimates his chance of passing the history test as 0.75 and of passing the Maths test as 0.9. Complete the tree diagram and then find the probability that Matt:

- a passes History and Maths _____ **E**
- b does not pass History or Maths _____ **W**
- c passes one of History or Maths _____ **O**
- d passes Maths but not History. _____ **V**



- 3 At a supermarket, a survey of 100 shoppers showed that 35 purchased bread, 32 purchased milk, 36 purchased fruit and 3 purchased all three items. It is also found that 5 purchased bread and milk, 16 purchased bread and fruit and 12 purchased milk and fruit. Find the probability that a shopper purchased:

- a only bread _____ **I**
- b only milk _____ **U**
- c milk and bread, but not fruit _____ **T**
- d bread or fruit _____ **L**
- e milk or fruit. _____ **Y**

